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Orthodontic device storage system

Abstract

A storage system for an orthodontic device is disclosed. The storage system includes a case having a first side, a second side engaged with the first side and forming an interior pocket between the first side and the second side. The storage system also includes a removal tool which includes a removal cloth engaged with the case. The removal tool includes a first pocket and a second pocket.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION(S) (1) This utility application claims priority to provisional application entitled Orthodontic Device Storage System, Ser. No. 63/453,914 filed Mar. 22, 2023 which is incorporated herein by reference.

TECHNICAL FIELD

(1) This disclosure relates to storage systems for orthodontic devices for use with devices including aligners, retainers, mouthguards dentures and other orthodontic devices.

BACKGROUND

(2) Many orthodontic devices exist and many of these orthodontic devices must be periodically removed from a user's mouth by the user. The user must then store the orthodontic device. A portable, reusable, and easy to use removal and storage system can help a user to remove and store their orthodontic devices.

Description

DESCRIPTION OF THE DRAWINGS

- (1) To allow for a more full understanding of the present disclosure, it will now be described by way of example, with reference to the accompanying drawings in which:
- (2) FIG. 1 illustrates a front perspective view of an exemplary orthodontic device storage system in use, in accordance with aspects of the disclosure;
- (3) FIG. 2 illustrates a back perspective view of the orthodontic device storage system of FIG. 1 in use, in accordance with aspects of the disclosure;
- (4) FIG. 3 illustrates a front view of the orthodontic device storage system of FIG. 1, in accordance with aspects of the disclosure;
- (5) FIG. 4 illustrates a back view of the orthodontic device storage system of FIG. 1, in accordance with aspects of the disclosure;
- (6) FIG. 5 illustrates a front view of the orthodontic device storage system of FIG. 1 with the removal tool inside the case, in accordance with aspects of the disclosure;
- (7) FIG. 6 illustrates a front view of another exemplary orthodontic device storage system, in accordance with aspects of the disclosure;
- (8) FIG. 7 illustrates a front perspective view of another exemplary orthodontic device storage system, in accordance with aspects of the disclosure;
- (9) FIG. 8 illustrates a front perspective view of another exemplary orthodontic device storage system, in accordance with aspects of the disclosure.

DETAILED DESCRIPTION

(10) While this invention is susceptible of embodiments in many different forms, there are shown in the drawings and will herein be described in detail example embodiments of the invention with the understanding that the present disclosure is to be considered as an exemplification of the principles of the invention and is not intended to limit the broad aspect of the invention to the embodiments illustrated. In the following description of various example structures according to the invention, reference is made to the accompanying drawings, which form a part hereof, and in which are shown by way of illustration various example devices, systems, and environments in which aspects of the invention may be practiced. It is to be understood that other specific arrangements of parts, example devices, systems, and environments may be utilized and structural and functional modifications may be made without departing from the scope of the present invention.

(11) Also, while the terms “top,” “bottom,” “front,” “back,” “side,” “distal,” and the like may be used in this specification to describe various example features and elements of the invention, these terms are used herein as a matter of convenience, e.g., based on the example orientations shown in the figures or the orientation during typical use. Nothing in this specification should be construed as requiring a specific three-dimensional orientation of structures in order to fall within the scope of this invention. Also, the reader is advised that the attached drawings are not necessarily drawn to scale.

(12) The following terms are used in this specification, and unless otherwise noted or clear from the context, these terms have the meanings provided below.

(13) “Plurality,” as used herein, indicates any number greater than one, either disjunctively or conjunctively, as necessary, up to an infinite number.

(14) In general, this disclosure relates to an orthodontic device storage system that is configured to store an orthodontic device and assist a user to more easily remove an orthodontic device. An orthodontic device can include any device a user may insert and remove from their mouth. This can include, for example, orthodontic aligners, dental prostheses, retainers, mouthguards, dentures, etc.

(15) FIGS. 1-5 illustrate a storage system **100** for an orthodontic device. As shown in FIGS. 1-5,

the storage system **100** can include a case **200**. The case as shown in FIGS. **1-5** is composed of a flexible material such as a textile, fabric, or cloth. However, as will be discussed below in other embodiments the case may be made of one or more rigid materials.

(16) As shown in FIGS. **1-5**, the case **200** includes a first side **202** and a second side **204** engaged with the first side **202**. The case **200** can also include a top end **203**, bottom end **205**, a first edge **207**, and a second edge **209**. The first side **202** and the second side **204** are engaged such that they form an interior pocket **206**. As shown in FIGS. **1-5** the first side **202** and the second side **204** can be engaged on the bottom end **205**, the first edge **207**, and the second edge **209** such that an opening **201** to the interior pocket **206** is formed in the top end **203**.

(17) As shown in FIGS. **1-5**, the storage system **100** can also include a removal tool **300**. The removal tool can include a removal cloth **302** having a front side **304**, and a back side **306**. The removal cloth **302** includes a first end **308** engaged with the case **200**, and a second end **310**. As shown the removal cloth **302** may be engaged on an inside portion of either the first side **202** or the second side **204** of the case **200**. In some embodiments the removal cloth **302** may be sewn to the case or may form a continuous portion of the first side **202** and/or the second side **204**. In other embodiments, however, the removal cloth **302** may be removably engaged with the case.

(18) As shown in FIGS. **1-4** the removal cloth may generally form a V-shape and may include a first pocket **312** forming a portion of the second end **204** and a second pocket **314** forming a portion of the second end **204**. The first pocket **312** may be sized to fit a user's first finger and surround the tip of a user's first finger and the second pocket **314** may be sized to fit a user's second finger and surround the tip of a user's second finger. As best shown in FIGS. **1** and **2**, a user can insert a finger into each of the first pocket **312** and the second pocket **314**. The user can pinch the orthodontic device between their fingers inserted in the first and second pockets **312**, **314** to remove the orthodontic device. Once the orthodontic device has been removed from the user's mouth, the user can insert the orthodontic device along with the removal cloth **300** into the interior pocket **206** of the case **200**. Advantageously the removal cloth **300** along with the orthodontic device can be completely stowed within the case **200**.

(19) To use the storage system **100** a user may adhere to the following steps: (1) the user opens the case **200** and extracts the removal tool **300**; (2) the user dons the removal tool **300** by inserting a finger into each of the first and second pockets **312**, **314**; (3) Using prehension between the first and second pockets **312**, **314**, the user grips the orthodontic device; (3) the removal tool **300** provides sufficient friction on the surface of orthodontic device to grasp, and in certain cases slightly torque, and remove the orthodontic device from the user's mouth; and (5) the user then may place the orthodontic device and the removal tool **300** into the case **200** in generally a single motion.

(20) The case **200** can also include a closure mechanism **220** to close the opening **201** such that the orthodontic device cannot accidentally fall out of the case **200**. The closure mechanism can include any suitable closure device. In some embodiments the closure mechanism can include, for example, elastic, a drawstring, a zipper, one or more magnets, one or more buttons, and a snap fit connection. The case **200** can also include an attachment device **230** that can be used to attach or hang the case to other items. The attachment device **230** could include, for example, a loop of material, a carabiner, a ring, a clip, or other suitable device

(21) The case **200** and the removal tool **300** can be made of any suitable material. For example, the case **200** and the removal tool **300** can be made of natural or synthetic materials and the case **200** and the removal tool **300** can be made of the same material or different materials. While the term removal cloth **302** has been used herein to describe a portion of the removal tool **300**, the removal tool **300** can be formed of other flexible materials that may not be cloth materials. Each of the case **200** and the removal tool **300** can be made of materials which may have antimicrobial materials or materials coated with antimicrobial agents. Further, in some embodiments, the case **200** and the removal tool **300** can be made of materials that allow for easy washing of the storage system **100**

such as by washing it in a dishwasher or laundry machine. In some embodiments the removal tool **300** can include additional treatments which may allow a user to more easily grip the orthodontic device such as by including gripping dots on the removal tool **300**.

(22) In certain embodiments the case **200** and removal tool **300** can have certain dimensions that may be advantageous to the usability of the storage system **100**. As best shown in FIGS. **3** and **5**. The removal tool **300** may have a length “A”. The case **300** may have a width “X” measured at the bottom end **205**, a height “Y”, and a width “Z” measured at the top end **203**. In some embodiments, the width “Z” measured at the top end **203** may be greater than the width “X” measured at the bottom end **205**. This may allow a user to more easily insert and remove the orthodontic device from the case **200**. Additionally, in some embodiments, the length of the removal tool “A” may be less than the height “Y” of the case **200** which may allow a user to more easily stow the removal tool **300** and the orthodontic device inside the case **200**. While the storage system **100** may have any suitable dimensions, in some embodiments the length “A” can be about 90 mm or in the range of about 70 mm-110 mm; the width “X” can be about 85 mm or in the range of about 65 mm-105 mm; the height “Y” can be about 110 mm or in the range of about 90 mm-130 mm, and the width “Z” can be about 90 mm or in the range of about 70 mm-110 mm.

(23) In some embodiments the storage system **100** can include other implements which may assist in removal of an orthodontic device. In some embodiments the removal tool **300** may include a removal instrument **320** extending from the removal cloth **302**. For example, as shown in FIG. **6**, the removal tool **300** can include a hook portion **322** extending from the removal cloth **302**. The hook portion **322** can be made of any suitable material including plastic. In such embodiments, a user could use the hook portion to assist in removal of the orthodontic device by inserting the hook portion **322** under a portion of the orthodontic device to loosen the orthodontic device from the user's mouth.

(24) As described and shown in FIGS. **1-6**, the case **300** can be formed of a flexible material. However, in other embodiments, the case **300** can be formed of a rigid material which may be more suitable for storage of certain orthodontic devices. As shown in FIG. **7** the case **400** can be formed of a rigid plastic material and as shown in FIG. **8** the case **500** can be formed of a rigid or firm material such as foam. The rigid cases **400**, **500** shown in FIGS. **7-8** can have similar components to the flexible case **200** shown above. The cases **400**, **500** can include a top portion **402**, **502** and a bottom portion **404**, **504**. The top portion **402**, **502** and the bottom portion **404**, **504** can be connected by a hinge portion **415**, **515**. Similar to the flexible case **200** described above, the top portion **402**, **502** and the bottom portion **404**, **504** form an interior pocket **406**, **506**. The removal tool **300** can be engaged with an interior portion of either the top portion **402**, **502** or the bottom portion **404**, **504**. Similarly, a user may store the orthodontic device and the removal tool **300** in the interior pocket **406**, **506** of the case **400**, **500**. Additionally, as described above, the case **400**, **500** can include a suitable closure mechanism such that the top portion **402**, **502** and the bottom portion **404**, **504** can be engaged together. The closure mechanism for cases **400**, **500** can include, for example, a zipper, one or more magnets, one or more buttons, or a snap fit connection

(25) The various embodiments described herein enable connection of an articulating boom arm. It is also understood that in other embodiments, the various devices, components, and features of the boom arm described herein may be constructed with similar structural and functional elements having different configurations, including different ornamental appearances. Still other benefits may be recognized by those skilled in the art. Accordingly, while the specific embodiments have been illustrated and described, numerous modifications come to mind without significantly departing from the spirit of the invention and the scope of protection is only limited by the scope of the accompanying claims.

Claims

1. A storage system for an orthodontic device, comprising: a case, the case comprising: a first side; a second side engaged with the first side and forming an interior pocket between the first side and the second side; a removal tool, the removal tool comprising: a removal cloth having a front side, a back side, a first end engaged with the case, and a second end; a first pocket forming a portion of the second end; a second pocket forming a portion of the second end; wherein the first pocket is sized to fit a user's first finger and wherein the second pocket is sized to fit a user's second finger; and wherein the removal cloth generally forms a V-shape.
2. The storage system for an orthodontic device of claim 1, wherein the case includes a closure mechanism.
3. The storage system for an orthodontic device of claim 2, wherein the closure mechanism is a drawstring.
4. The storage system for an orthodontic device of claim 2, wherein the closure mechanism is a zipper.
5. The storage system for an orthodontic device of claim 2, wherein the closure mechanism is magnetic.
6. The storage system for an orthodontic device of claim 2, wherein the closure mechanism includes one or more buttons.
7. The storage system for an orthodontic device of claim 2, wherein the closure mechanism is a snap fit connection.
8. The storage system for an orthodontic device of claim 1, further comprising a removal instrument extending from the removal cloth.
9. The storage system for an orthodontic device of claim 8, wherein the removal instrument comprises a hook portion.
10. A storage system for an orthodontic device, comprising: a case, the case comprising: a first side; a second side engaged with the first side and forming an interior pocket between the first side and the second side; a removal tool, the removal tool comprising: a removal cloth having a front side, a back side, a first end engaged with the case, and a second end; a first pocket forming a portion of the second end; a second pocket forming a portion of the second end; wherein the first pocket is sized to fit a user's first finger and surround a tip of the user's first finger and wherein the second pocket is sized to fit a user's second finger and surround the tip of a user's second finger; wherein the removal cloth generally forms a V-shape; and wherein the case includes a closure mechanism.
11. The storage system for an orthodontic device of claim 10, wherein the closure mechanism is a drawstring.
12. The storage system for an orthodontic device of claim 11, wherein the closure mechanism is a zipper.
13. The storage system for an orthodontic device of claim 11, wherein the closure mechanism is magnetic.
14. The storage system for an orthodontic device of claim 11, wherein the closure mechanism includes one or more buttons.
15. The storage system for an orthodontic device of claim 11, wherein the closure mechanism is a snap fit connection.
16. The storage system for an orthodontic device of claim 10, further comprising a removal instrument extending from the removal cloth.
17. The storage system for an orthodontic device of claim 16, wherein the removal instrument comprises a hook portion.
18. The storage system for an orthodontic device of claim 10, wherein case is formed of a rigid material.

19. The storage system for an orthodontic device of claim 10, wherein the removal tool is removably engaged with the case.
